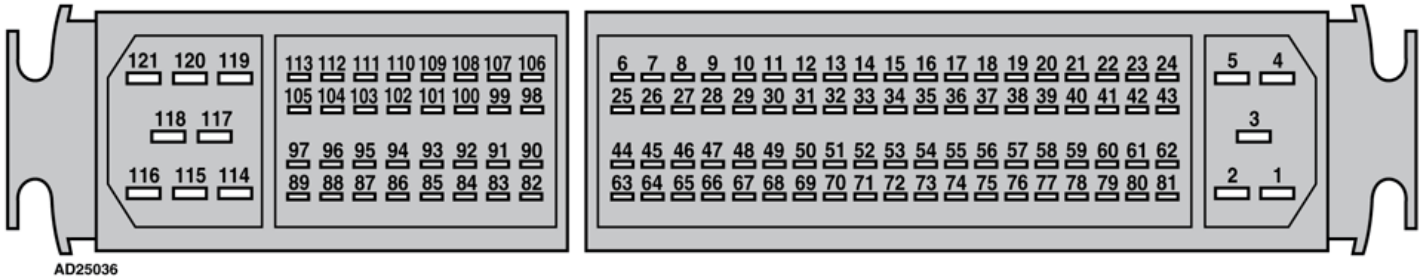


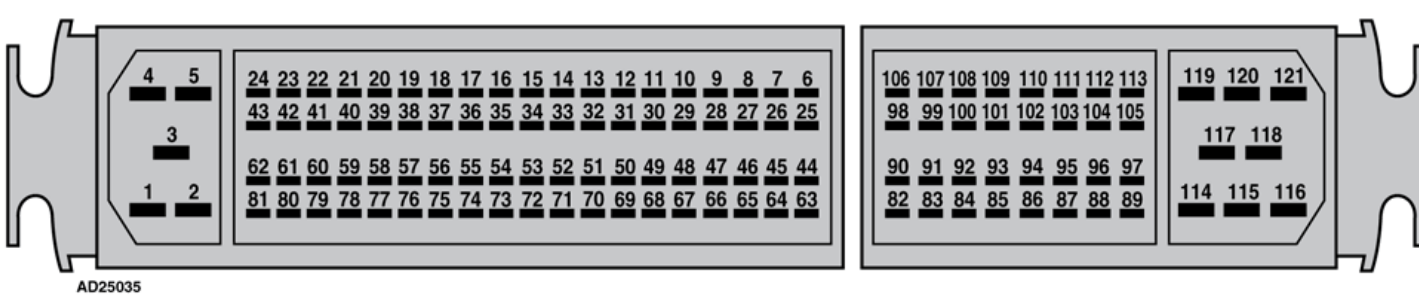
Pin Data

Audi TT
BAM/1.8 (8N3/8N9) (98-06)

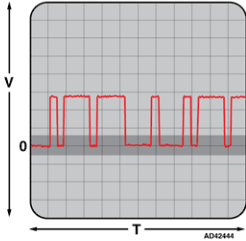
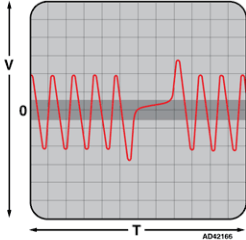
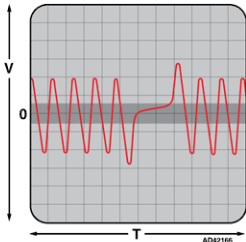
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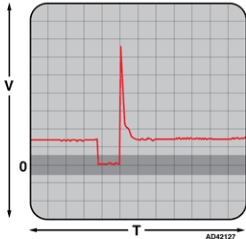
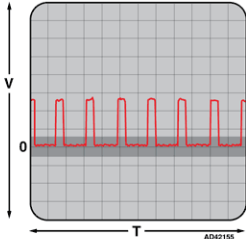
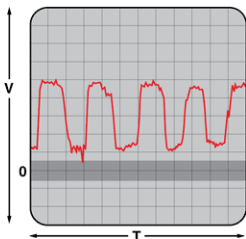


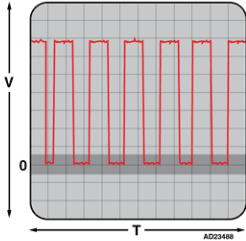
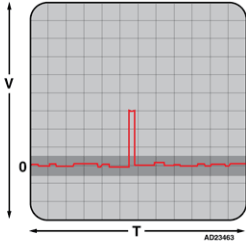
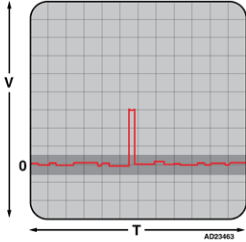
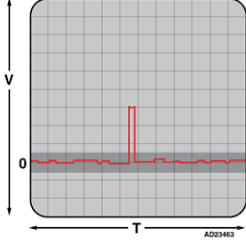
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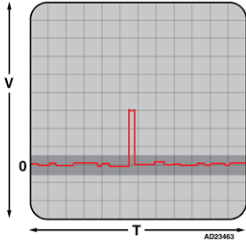
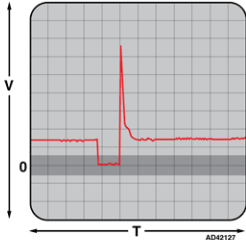
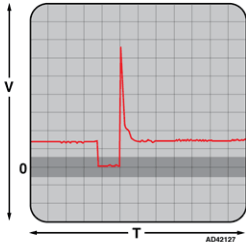
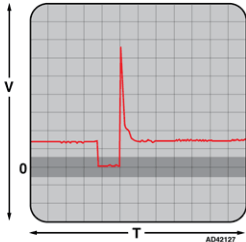
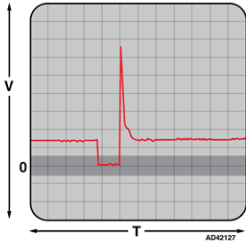


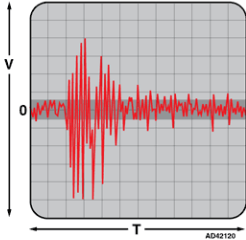
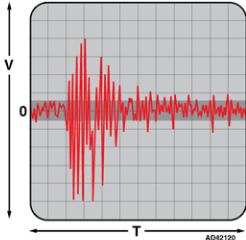
Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
Accelerator pedal position (APP) sensor	33		Ignition ON	0V		
Accelerator pedal position (APP) sensor	34		Ignition ON - accelerator pedal released	0,4V		
Accelerator pedal position (APP) sensor	34		Ignition ON - accelerator pedal depressed	2V		
Accelerator pedal position (APP) sensor	35		Ignition ON - accelerator pedal released	0,7V		
Accelerator pedal position (APP) sensor	35		Ignition ON - accelerator pedal depressed	4,1V		
Accelerator pedal position (APP) sensor	36		Ignition ON	0V		
Accelerator pedal position (APP) sensor	72		Ignition ON	5V		
Accelerator pedal position (APP) sensor	73		Ignition ON	5V		
Air conditioning	40			Connected pin - no test data available or random digital signal		
Air conditioning	41			Connected pin - no test data available or random digital signal		
Air conditioning - 2001-	37			Connected pin - no test data available or random digital signal		
Alternator - some models	28		Engine idling		5 V/50 ms	
Battery	62		Ignition OFF	11-14V		
Brake pedal position (BPP) switch 1	56		Ignition ON - brake pedal released	0V		

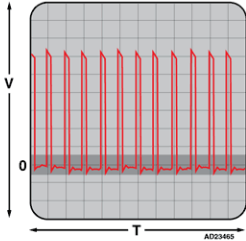
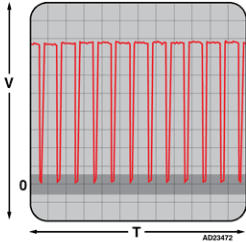
Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
<u>Brake pedal position (BPP) switch 1</u>	56	⬅	Ignition ON - brake pedal depressed	11-14V		
<u>Brake pedal position (BPP) switch 2</u>	55	⬅	Ignition ON - brake pedal released	11-14V		
<u>Brake pedal position (BPP) switch 2</u>	55	⬅	Ignition ON - brake pedal depressed	0V		
<u>Camshaft position (CMP) actuator - some models</u>	115	↔	Ignition ON	11-14V briefly then 0-1V		
<u>Camshaft position (CMP) actuator - some models</u>	115	↔	Engine idling	11-14V		
<u>Camshaft position (CMP) sensor</u>	86	⬅	Ignition ON	0 or 11-14V		
<u>Camshaft position (CMP) sensor</u>	86	⬅	Engine idling		5 V/20 ms	
<u>Camshaft position (CMP) sensor</u>	98	⇒	Ignition ON	5V		
<u>Camshaft position (CMP) sensor</u>	108	↗	Ignition ON	0V		
CAN data bus - high	60	↔		Connected pin - no test data available or random digital signal		
CAN data bus - low	58	↔		Connected pin - no test data available or random digital signal		
<u>Clutch pedal position (CPP) switch</u>	39	⬅	Ignition ON - clutch pedal released	11-14V		
<u>Clutch pedal position (CPP) switch</u>	39	⬅	Ignition ON - clutch pedal depressed	0V		
<u>Crankshaft position (CKP) sensor</u>	82	⬅	Engine idling	3,1V ac		
<u>Crankshaft position (CKP) sensor</u>	82	⬅	Engine idling		2 V/2 ms	
<u>Crankshaft position (CKP) sensor</u>	90	⬅	Engine idling	3,1V ac		
<u>Crankshaft position (CKP) sensor</u>	90	⬅	Engine idling		2 V/2 ms	
						Reversed
<u>Crankshaft position (CKP) sensor</u>	108	↗	Ignition ON	0V		
<u>Cruise control master switch</u>	38			Connected pin - no test data available or random digital signal		
<u>Cruise control master switch</u>	57			Connected pin - no test data available or random digital signal		
<u>Cruise control selector switch</u>	38			Connected pin - no test data available or random digital signal		
<u>Cruise control selector switch</u>	75			Connected pin - no test data available or random digital signal		

Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
<u>Cruise control selector switch</u>	76			Connected pin - no test data available or random digital signal		
Earth	1		Ignition ON	0V		
Earth	2		Ignition ON	0V		
<u>Engine control (EC) relay - 2000.</u>	21		Ignition ON	0-1V		
<u>Engine control (EC) relay - 2000.</u>	21		Engine idling	0-1V		
<u>Engine control (EC) relay - 2000.</u>	121		Ignition ON	11-14V		
<u>Engine coolant blower motor/AC compressor control module - AIQ/APP/APX</u>	61			Connected pin - no test data available or random digital signal		
<u>Engine coolant temperature (ECT) sensor</u>	93		Ignition ON - coolant temp. 20°C	2,2V		
<u>Engine coolant temperature (ECT) sensor</u>	93		Ignition ON - coolant temp. 80°C	0,4V		
<u>Engine coolant temperature (ECT) sensor</u>	108		Ignition ON	0V		
<u>Engine malfunction indicator lamp (MIL) - without CAN data bus</u>	47		Ignition ON - MIL ON	0-1V		
<u>Engine malfunction indicator lamp (MIL) - without CAN data bus</u>	47		Engine idling - MIL OFF	11-14V		
<u>Evaporative emission (EVAP) canister purge valve</u>	64		Ignition ON	11-14V briefly then 0-1V		
<u>Evaporative emission (EVAP) canister purge valve</u>	64		Engine running - engine hot - valve operating		10 V/5 ms	
<u>Exhaust gas temperature sensor - BAM</u>	61			Connected pin - no test data available or random digital signal		
<u>Exhaust gas temperature sensor - BAM → 2001</u>	50		Ignition ON	0V		
<u>Fuel pump (FP) relay</u>	65		Ignition ON	0-1V briefly then 11-14V		
<u>Fuel pump (FP) relay</u>	65		Engine idling	0-1V		
<u>Heated oxygen sensor (HO2S) 1 - AIQ/APP/APX</u>	5				5 V/50 ms	
<u>Heated oxygen sensor (HO2S) 1 - AIQ/APP/APX</u>	51		Engine idling	0V		
<u>Heated oxygen sensor (HO2S) 1 - AIQ/APP/APX</u>	70		Engine idling	0,1-1V fluctuating	0,2 V/1 sec.	

Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	5				2 V/0,2 sec.	
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	51		Engine idling	2,5V		
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	51 (70)		Engine idling	0,45V		
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	52		Engine idling	2,34-2,55V		
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	70		Engine idling	2,9V		
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	70 (51)		Engine idling	0,45V		
Heated oxygen sensor (HO2S) 1 - ARY/AUM/AUQ/BAM	71		Engine idling	2,34-2,55V		
Heated oxygen sensor (HO2S) 2 - APP/ARY/AUM/AUQ/BAM	63		Ignition ON	11-14V briefly then 0-1V		
Heated oxygen sensor (HO2S) 2 - APP/ARY/AUM/AUQ/BAM	63		Engine idling	0-1V		
Heated oxygen sensor (HO2S) 2 - APP/ARY/AUM/AUQ/BAM	68		Ignition ON	0V		
Heated oxygen sensor (HO2S) 2 - APP/ARY/AUM/AUQ/BAM	69		Engine idling - engine hot	0,6V		
Ignition coil 1	102		Ignition ON	0V		
Ignition coil 1	102		Engine idling		2 V/20 ms	
Ignition coil 2	95		Ignition ON	0V		
Ignition coil 2	95		Engine idling		2 V/20 ms	
Ignition coil 3	103		Ignition ON	0V		
Ignition coil 3	103		Engine idling		2 V/20 ms	
Ignition coil 4	94		Ignition ON	0V		

Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
<u>Ignition coil 4</u>	94	⇔	Engine idling		2 V/20 ms	
<u>Ignition switch</u>	3	⬅	Ignition OFF	0V		
<u>Ignition switch</u>	3	⬅	Ignition ON	11-14V		
<u>Injector 1</u>	96	⇨	Ignition ON	11-14V briefly then 0-1V		
<u>Injector 1</u>	96	⇨	Engine idling	2,5ms	10 V/2 ms	
<u>Injector 2</u>	89	⇨	Ignition ON	11-14V briefly then 0-1V		
<u>Injector 2</u>	89	⇨	Engine idling	2,5ms	10 V/2 ms	
<u>Injector 3</u>	97	⇨	Ignition ON	11-14V briefly then 0-1V		
<u>Injector 3</u>	97	⇨	Engine idling	2,5ms	10 V/2 ms	
<u>Injector 4</u>	88	⇨	Ignition ON	11-14V briefly then 0-1V		
<u>Injector 4</u>	88	⇨	Engine idling	2,5ms	10 V/2 ms	
<u>Instrumentation control module - diagnosis signal</u>	43			Connected pin - no test data available or random digital signal		
<u>Instrumentation control module - engine RPM signal-2001</u>	37	⇔	Engine idling	28Hz		
<u>Instrumentation control module - vehicle speed signal</u>	54	⬅		Connected pin - no test data available or random digital signal		

Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
<u>Instrumentation control module - some models</u>	30			Connected pin - no test data available or random digital signal		
<u>Instrumentation control module - some models</u>	81			Connected pin - no test data available or random digital signal		
<u>Intake air temperature (IAT) sensor</u>	85	←■	Ignition ON - air temp. 20°C	2,1V		
<u>Intake air temperature (IAT) sensor</u>	108	↯	Ignition ON	0V		
<u>Knock sensor (KS) 1</u>	99	↯	Engine idling	0V		
<u>Knock sensor (KS) 1</u>	106	←■	Engine idling - accelerate briefly		50 mV/1 ms	
<u>Knock sensor (KS) 1</u>	108	↯	Ignition ON	0V		
<u>Knock sensor (KS) 2</u>	99	↯	Engine idling	0V		
<u>Knock sensor (KS) 2</u>	107	←■	Engine idling - accelerate briefly		50 mV/1 ms	
<u>Knock sensor (KS) 2</u>	108	↯	Ignition ON	0V		
<u>Mass air flow (MAF) sensor</u>	27	↯	Ignition ON	0V		
<u>Mass air flow (MAF) sensor</u>	29	←■	Engine idling	1,4V		
<u>Mass air flow (MAF) sensor</u>	53	⇔	Ignition ON	5V		
<u>Power steering pressure (PSP) switch - some models</u>	49	←■	Engine idling - steering wheel not turned	11-14V		
<u>Power steering pressure (PSP) switch - some models</u>	49	←■	Engine idling - steering wheel turned	0V		
<u>Power steering pressure (PSP) switch - some models</u>	50	↯	Ignition ON	0V		
<u>Secondary air injection (AIR) pump relay - except AIQ</u>	66	↯	Ignition ON	11-14V briefly then 0-1V		
<u>Secondary air injection (AIR) pump relay - except AIQ</u>	66	↯	Engine running - pump OFF	11-14V		
<u>Secondary air injection (AIR) pump relay - except AIQ</u>	66	↯	Engine running - pump ON	0-1V		
<u>Secondary air injection (AIR) solenoid - except AIQ</u>	9	↯	Ignition ON	11-14V briefly then 0-1V		
<u>Secondary air injection (AIR) solenoid - except AIQ</u>	9	↯	Engine running - solenoid OFF	11-14V		
<u>Secondary air injection (AIR) solenoid - except AIQ</u>	9	↯	Engine running - solenoid ON	0-1V		
<u>Supplementary restraint system (SRS) control module</u>	67			Connected pin - no test data available or random digital signal		
<u>Throttle control system warning lamp - without CAN data bus</u>	48	↯	Ignition ON - warning lamp ON	0-1V		
<u>Throttle control system warning lamp - without CAN data bus</u>	48	↯	Engine idling - warning lamp OFF	11-14V		
<u>Throttle motor</u>	117	⇔	Ignition ON	11-14V for 30seconds then 3V		

Component/circuit description	ECM pin	Signal	Condition	Typical value	Oscilloscope setting	Wave form
<u>Throttle motor</u>	117	⇔	Ignition ON - accelerator pedal released		2 V/0,5 ms	
<u>Throttle motor</u>	117	⇔	Ignition ON - accelerator pedal depressed		2 V/0,5 ms	
<u>Throttle motor</u>	118	⇔	Ignition ON	11-14V		
<u>Throttle motor</u>	118	⇔	Engine idling	11-14V		
<u>Throttle motor position sensor</u>	83	⇔	Ignition ON	5V		
<u>Throttle motor position sensor</u>	84	←■	Ignition ON - accelerator pedal released	4,3V		
<u>Throttle motor position sensor</u>	84	←■	Ignition ON - accelerator pedal depressed	0,7V		
<u>Throttle motor position sensor</u>	91	⌈	Ignition ON	0V		
<u>Throttle motor position sensor</u>	92	←■	Ignition ON - accelerator pedal released	0,7V		
<u>Throttle motor position sensor</u>	92	←■	Ignition ON - accelerator pedal depressed	4,3V		
<u>Turbocharger (TC) boost pressure sensor</u>	98	⇔	Ignition ON	5V		
<u>Turbocharger (TC) boost pressure sensor</u>	101	←■	Ignition ON	1,9V		
<u>Turbocharger (TC) boost pressure sensor</u>	101	←■	Engine idling	1,9V		
<u>Turbocharger (TC) boost pressure sensor</u>	108	⌈	Ignition ON	0V		
<u>Turbocharger (TC) intake divert valve</u>	105	⌈⌋	Ignition ON	11-14V briefly then 0-1V		
<u>Turbocharger (TC) intake divert valve</u>	105	⌈⌋	Engine idling	11-14V		
<u>Turbocharger (TC) wastegate regulating valve</u>	104	⌈⌋	Ignition ON	11-14V briefly then 0-1V		
<u>Turbocharger (TC) wastegate regulating valve</u>	104	⌈⌋	Engine idling	11-14V		